

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

Mohammed Mahmoud Elnaggar

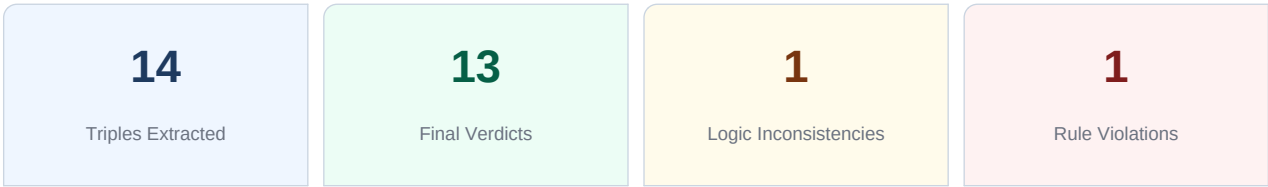
Generated on: 2026-04-30 19:38 UTC

REPORT OVERVIEW & METHODOLOGY

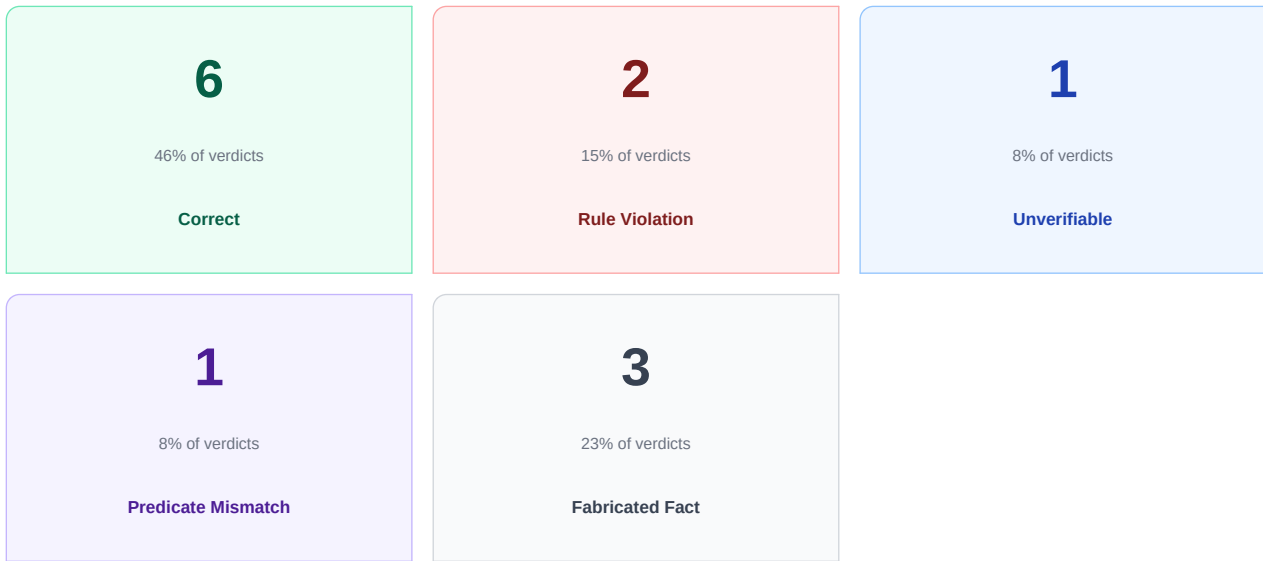
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	300	HAS_INTEREST_RATE	22.94		The percentage charged on a loan or paid on an investment.
2	305	HAS_INTEREST_RATE	Not in context		The percentage charged on a loan or paid on an investment.
3	305	HAS_APPROVAL_ODDS	8		The likelihood or score indicating the chance of approval.
4	310	HAS_APPROVAL_ODDS	10		The likelihood or score indicating the chance of approval.
5	310	HAS_INTEREST_RATE	23		The percentage charged on a loan or paid on an investment.
6	313	HAS_INTEREST_RATE	Not in context		The percentage charged on a loan or paid on an investment.
7	300	HAS_COLLATERAL_TYPE	real_estate		The category or nature of the asset pledged as security for a loan.
8	301	HAS_COLLATERAL_TYPE	real_estate		The category or nature of the asset pledged as security for a loan.
9	600	HAS_COLLATERAL_TYPE	vehicle		The category or nature of the asset pledged as security for a loan.
10	700	HAS_COLLATERAL_TYPE	Not in context		The category or nature of the asset pledged as security for a loan.
11	303	HAS_INTEREST_RATE	22.94		The percentage charged on a loan or paid on an investment.
12	304	HAS_INTEREST_RATE	22.95		The percentage charged on a loan or paid on an investment.
13	307	HAS_INTEREST_RATE	Not in context		The percentage charged on a loan or paid on an investment.
14	France	HAS_CAPITAL	Paris		The primary city or town associated with a country or region where the government is located.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
300	HAS_INTEREST_RATE	22.94	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
305	HAS_INTEREST_RATE	Not in context	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
305	HAS_APPROVAL_ODDS	8	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
310	HAS_APPROVAL_ODDS	10	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
310	HAS_INTEREST_RATE	23	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
313	HAS_INTEREST_RATE	Not in context	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
300	HAS_COLLATERAL_TYPE	real_estate	HAS_COLLATERAL_TYPE	HIGH
301	HAS_COLLATERAL_TYPE	real_estate	HAS_COLLATERAL_TYPE	HIGH
600	HAS_COLLATERAL_TYPE	vehicle	HAS_COLLATERAL_TYPE	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Conf.
700	HAS_COLLATERAL_TYPE	Not in context	HAS_COLLATERAL_TYPE	HIGH
303	HAS_INTEREST_RATE	22.94	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
304	HAS_INTEREST_RATE	22.95	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
307	HAS_INTEREST_RATE	Not in context	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
France	HAS_CAPITAL	Paris	HAS_CAPITAL	LOW

FACTUALLY INCONSISTENT TRIPLES

All triples labelled `factually_inconsistent` from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
<code>factually_inconsistent</code>	<code>RULE_VIOLATION</code>	310	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	10	<code>c1_approval_odds</code>	Credit score 310 requires object to semantically contain '8', but object is '10'.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
301	HAS_COLLATERAL_TYPE	real_estate	Fabricated Fact	<code>factually_inconsistent</code>	<code>FABRICATED_FACT</code>	NONE	SUBJECT	0.90	Object+predicate exist in DB but no matching subject found; subject is fabricated.
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	22.94	Fabricated Fact	<code>factually_inconsistent</code>	<code>FABRICATED_FACT</code>	NONE	SUBJECT	0.90	Object+predicate exist in DB but no matching subject found; subject is fabricated.
France	HAS_CAPITAL	Paris	Unverifiable	<code>unverifiable</code>	—	NONE	—	0.00	Predicate is not present in the DB schema; triple cannot be verified.
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	22.94	Correct	<code>factually_consistent</code>	—	STRICT	—	1.00	The object '22.94' exactly matches the DB candidate for subject '300' and predicate 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE': <i>Ev: subject: 300, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.94</i>

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
305	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	Not in context	Rule Violation	factually_inconsistent	RULE_VIOLATION	NONE	OBJECT	●●●●●●●● ●●● 1.00	The DB candidate provides a concrete value '22.61' but the object states absence ('Not in context'), which contradicts the stored value. <i>Ev: subject: 305, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.61</i>
305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	8	Correct	factually_consistent	—	STRICT	—	●●●●●●●● ●●● 1.00	The object '8' exactly matches the DB candidate for subject '305' and predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD'. <i>Ev: subject: 305, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 8</i>
310	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	23	Rule Violation	factually_inconsistent	RULE_VIOLATION	NONE	OBJECT	●●●●●●●● ●●● 1.00	The object '23' does not match the DB candidate value '22.19' for subject '310' and predicate 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE'. <i>Ev: subject: 310, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.19</i>
313	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	Not in context	Correct	factually_consistent	—	NONE	—	●●●●●●●● ●●■ 0.90	The object 'Not in context' signals the absence of information, which is appropriate since no DB records were found for this predicate.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
300	HAS_COLLATERAL_TYPE	real_estate	Correct	factually_consistent	—	SEMANANTIC	—	●●●●●●●● ●●● 1.00	The object 'real_estate' matches the DB candidate for subject '300' and predicate 'HAS_COLLATERAL_TYPE'. <i>Ev: subject: 300, predicate: HAS_COLLATERAL_TYPE, object: real_estate</i>
600	HAS_COLLATERAL_TYPE	vehicle	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	NONE	PREDICATE	●●●●●●●● ●●■ 0.90	The object 'vehicle' asserts a concrete value, but this predicate is not a valid relation for subject '600' in the DB.
700	HAS_COLLATERAL_TYPE	Not in context	Correct	factually_consistent	—	NONE	—	●●●●●●●● ●●■ 0.90	The object 'Not in context' signals absence of information and is valid for Lane B when no DB record exists.
304	INTEREST_RATE_CHARGED_AS_ANNUALBORROWING_COST_PERCENTAGE	22.95	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● ●●■ 0.90	The object '22.95' asserts a concrete value unsupported by any DB evidence for this predicate and subject.
307	INTEREST_RATE_CHARGED_AS_ANNUALBORROWING_COST_PERCENTAGE	Not in context	Correct	factually_consistent	—	NONE	—	●●●●●●●● ●●■ 0.90	The object 'Not in context' signals absence of information, which is appropriate since no DB records were found for this predicate.

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

Label	Type	Severity	Explanation	Triples Involved
logically_inconsistent	MONOTONICITY VIOLATION	MEDIUM	A higher credit score (310) has a strictly higher interest rate (23%) than a lower credit score (300 with 22.94%), violating the DECREASING requirement for INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE.	300 -> INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE -> 22.94 310 -> INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE -> 23

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship that is in db but the subject does not supported by db.